Estimating SET100 Realized Beta with Long Short-term Memory Model



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การประมาณค่าเบต้าที่เกิดขึ้นจริงของหุ้นในดัชนี SET100 ด้วยแบบจำลองหน่วยความจำระยะสั้นแบบยาว

นายอิสระ คามบุตร

สารนิพนธ์นี้เป็นส่วนหนึ่งของการศึกษาตามหลักสูตรปริญญาวิทยาศาสตรมหาบัณฑิต
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อิสระ คามบุตร : การประมาณค่าเบต้าที่เกิดขึ้นจริงของหุ้นในดัชนี SET100 ด้วยแบบจำลองหน่วยความจำระยะสั้นแบบยาว (Estimating SET100 Realized Beta with Long Short-term Memory Model) อาจารย์ที่ปรึกษาหลัก: รองศาสตราจารย์ ดร.อนิรุต พิเสฏฐศลาศัย

การศึกษานี้ต้องการตอบคำถามว่า แบบจำลองหน่วยความจำระยะสั้นแบบยาว ซึ่งฝึกด้วยตัวแปรพยากรณ์สิบตัวทั้งในระดับบริษัทและระดับเศรษฐกิจมหภาค จะพยากรณ์ค่าเบต้าที่เกิดขึ้นจริงล่วงหน้าหนึ่งปีของหุ้นกลุ่ม SET100 ได้แม่นยำกว่าวิธีที่นักปฏิบัตินิยมใช้หรือไม่ ได้แก่ การถดถอยแบบหน้าต่างเลื่อนด้วยข้อมูลรายวันย้อนหลังหนึ่งปี และด้วยข้อมูลรายเดือนย้อนหลังห้าปี โดยทดสอบด้วยการพยากรณ์นอกกลุ่มตัวอย่างรายเดือนตั้งแต่ปี 2015 ถึง 2024 และประเมินผลแบบถ่วงน้ำหนักตามมูลค่าตลาด ผลการศึกษาพบว่า แบบจำลองหน่วยความจำระยะสั้นแบบยาวพยากรณ์เบต้าได้แม่นยำกว่าการถดถอยรายเดือนห้าปีในหุ้นทุกกลุ่มขนาด ทั้งในช่วงก่อนและหลังโควิด-19 แต่เมื่อเทียบกับการถดถอยรายวันหนึ่งปี แบบจำลองนี้กลับไม่ได้เหนือกว่า ทั้งสองวิธีให้ความแม่นยำไม่ต่างกันอย่างมีนัยสำคัญทางสถิติเมื่อวัดในภาพรวมแบบถ่วงน้ำหนักตามมูลค่าตลาด และผลนี้คงเดิมทั้งก่อนและหลังโควิด-19 ซึ่งสวนทางกับความคาดหวังทั่วไปที่ว่าแบบจำลองการเรียนรู้เชิงลึกที่ยืดหยุ่นกว่าน่าจะชนะการถดถอยอย่างง่ายได้ในทุกกรณี

คำอธิบายที่เป็นไปได้คือ การถดถอยรายวันหนึ่งปีประมาณค่าเบต้าของหุ้นขนาดใหญ่ที่มีสภาพคล่องสูงได้ดีอยู่แล้ว แบบจำลองที่ซับซ้อนกว่าจึงแทบไม่มีช่องว่างให้เพิ่มความแม่นยำในหุ้นกลุ่มนี้ ซึ่งเป็นกลุ่มที่มีน้ำหนักมากที่สุดในการประเมินแบบถ่วงน้ำหนักตามมูลค่าตลาด จุดแข็งของแบบจำลองหน่วยความจำระยะสั้นแบบยาวจึงอยู่ที่หุ้นขนาดเล็กและขนาดกลางแทน โดยเฉพาะกลุ่มขนาดกลางที่ผลต่างมีนัยสำคัญทางสถิติ นอกจากนี้ การพยากรณ์ของแบบจำลองอาศัยค่าเบต้าในอดีตช่วงล่าสุดเป็นหลัก มากกว่าจะอาศัยตัวแปรพยากรณ์ตัวอื่น ๆ อย่างไรก็ตาม ความได้เปรียบนี้ปรากฏเฉพาะบางส่วนของตลาดเท่านั้น และในหุ้นขนาดเล็กจะเห็นชัดที่สุดในช่วงก่อนการระบาดของโควิด-19 ดังนั้นต้นทุนที่เพิ่มขึ้นจากการใช้แบบจำลองหน่วยความจำระยะสั้นแบบยาวจึงคุ้มค่าเฉพาะกับหุ้นกลุ่มขนาดเล็กและขนาดกลางเป็นหลัก

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This study examines whether a Long Short-Term Memory (LSTM) network, trained on ten firm-level and macroeconomic predictors, can forecast the one-year-ahead realized beta of SET100 stocks more accurately than the one-year daily and five-year monthly rolling regressions that practitioners rely on, using monthly out-of-sample forecasts from 2015 to 2024 evaluated on a value-weighted basis. The result suggests that the LSTM forecasts beta more accurately than the five-year monthly regression in every size group and in both COVID-19 regimes. However, it does not dominate the one-year daily regression. In fact, the two models are statistically equal on the value-weighted aggregate, and this parity holds both before and after the COVID-19 break, contradicting a common expectation that a flexible deep-learning model should beat simple regressions across the board.

A possible explanation for this finding is that a one-year daily regression already estimates the betas of large and liquid stocks with little bias, so little room is left for a richer model to add value among the stocks that dominate the value-weighted measure. The network instead earns its advantage among small and mid-sized stocks, significantly so in the mid group, and its forecasts depend mainly on the recent historical betas rather than on the wider predictor set. Finally, these gains are observed in some, but not all, segments of the market, as the advantage among smaller stocks is clearest before the pandemic, so the additional cost of an LSTM is justified mainly for the small and mid-cap segment.

Field of Study	Finance	Student's Signature .....
Academic Year	2025	Advisor's Signature .....
		Co-advisor's Signature .....
		Co-advisor's Signature .....

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Lastly, I want to thank you, the reader, for dedicating your time to my work. Whether you came here out of genuine interest, or because random chance led you here, I hope my work rewards the visit. And if you have read this far, the rest of the paper should hold no fear for you.

Isara Khambut

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Chapter 1

Introduction

The capital asset pricing model of Sharpe (1964) describes a linear relationship between an asset's expected return and its beta. Beta measures how much the asset moves with the market, and it shows the systematic risk that cannot be diversified away. Beta is important in finance because it is used to estimate the cost of equity, to build and hedge portfolios, and to manage risk. However, beta is hard to use in practice: it cannot be observed directly, and it changes over time (Drobetz et al., 2025).

A common way to estimate beta is to run a regression over a rolling window of past returns, using either a long window of monthly data or a short window of daily data. This approach faces a bias-variance trade-off: a short window reacts quickly to change but gives noisy estimates, while a long window is more stable but slower to adapt. To reduce this problem, researchers have added some adjustments. Vasicek (1973) and Blume (1975) shrink noisy estimates toward a common central value, and Cosemans et al. (2016) combine past returns with firm fundamentals.

More recently, machine learning has improved asset-pricing predictions. For example, Gu et al. (2020) compare several methods and show that neural networks and tree-based models predict returns and risk premia more accurately than linear regressions. They do this mainly by capturing nonlinear effects and interactions between predictors that linear models miss. Drobetz et al. (2025) apply machine learning to beta estimation and show that these models can match or beat standard benchmarks. The Long Short-Term Memory (LSTM) network of Hochreiter and Schmidhuber (1997) is built for sequential data and can keep information over long periods without the vanishing-gradient problem. Fischer and Krauss (2018) apply the LSTM to financial time series and find that it works better than models without memory.

Almost all of this evidence comes from developed markets, especially the United States. Emerging markets are studied much less, even though they differ in ways that matter for beta estimation, such as thinner trading, higher concentration in a few large stocks, and more frequent regime shifts. The Stock Exchange of Thailand is a clear example. Prukumpai (2015) models the betas of eight in-

dustry portfolios with a two-regime Markov-switching model and finds that they are time-varying and regime-dependent. However, this evidence covers only industry portfolios and uses parametric models. It is therefore still unclear whether the gains that machine learning brings in large and liquid markets also hold in a smaller and more concentrated emerging market.

This study looks at that question for the SET100 index. Its aim is to test whether an LSTM model can forecast the one-year-ahead realized beta of SET100 stocks more accurately than rolling-regression benchmarks that use past returns alone. The model uses ten firm-level and macroeconomic predictors, together with a walk-forward design that produces monthly out-of-sample forecasts from 2015 to 2024, and it is evaluated on a value-weighted out-of-sample basis. The study tests two hypotheses. The first is that the LSTM forecasts beta more accurately than one-year daily and five-year monthly rolling regressions. The second is that its relative accuracy is stable across the COVID-19 break. Two further analyses then deepen the comparison: one traces where the advantage lies across firm size, and the other asks which predictors matter most for the forecasts.

The study makes three contributions. For the literature, it provides what appears to be the first application of an LSTM to realized-beta forecasting in the Thai market, which extends the machine-learning beta literature beyond developed economies. For method, it tests the model against two strong rolling regressions using a value-weighted, out-of-sample measure, so that larger stocks count for more instead of every stock being treated equally. For practice, it uses a predictor-importance analysis to show which inputs the forecasts rely on most, which gives useful guidance for risk management.

Chapter 2

Literature Review

2.1 The Capital Asset Pricing Model and Market Beta

Market beta measures an asset's systematic risk, which is the part of its risk that comes from moving together with the market and cannot be diversified away. It is the main quantity in the capital asset pricing model of Sharpe (1964), which predicts a linear relationship between an asset's expected return and its beta. Early empirical tests generally found a positive relationship between beta and average returns (Black et al., 1972; Fama and MacBeth, 1973). Beta is therefore a basic input for estimating the cost of equity, building portfolios, and managing risk. However, it cannot be observed directly, so it has to be estimated from returns, and the way it is estimated has a large effect on the result.

2.2 Traditional Beta Estimation and Its Limits

The most common way to estimate beta is the rolling-window regression, which estimates beta over a moving window of recent returns, for example five years of monthly data (Fama and MacBeth, 1973) or one year of daily data (Andersen et al., 2006). This design faces a bias-variance trade-off: a short window adapts quickly to change but gives noisy estimates, while a long window is more stable but slow to reflect new information. Rolling estimates are also affected by outliers and by non-synchronous or thin trading, which is a particular problem in smaller markets. Several methods try to reduce this noise. Vasicek (1973) and Blume (1975) shrink raw estimates toward a common central value, and Cosemans et al. (2016) combine past returns with firm fundamentals to improve beta forecasts across stocks. Even with these adjustments, Hollstein et al. (2019) show that beta is still estimated with a lot of uncertainty, and that the forecast errors are systematically larger for some kinds of stocks, such as small, illiquid, and volatile ones. These methods improve a backward-looking estimate, but they still depend on a fixed window and react to change only after it appears in the data.

2.3 The Time Variation of Beta

A deeper problem is that beta changes over time. The original model treats beta as constant, but many studies show that systematic risk varies over time with economic conditions and firm characteristics. Jagannathan and Wang (1996) develop a conditional CAPM in which beta moves with the business cycle, and they show that it explains average returns across stocks much better than the static model. Becker et al. (2021) find long-memory behaviour in beta, which means that its changes are persistent rather than short-lived. Realized beta makes it possible to measure these changes directly. Andersen et al. (2006) show that realized beta, built from finely sampled returns over a fixed horizon, is a consistent measure of a stock's average systematic risk over that horizon. Later studies evaluate beta forecasts against it (Becker et al., 2021; Drobetz et al., 2025). This study uses realized beta as its forecasting target. Because beta really does change over time, the rolling and shrinkage estimators in the previous section are slow: they follow change only with a delay and do not model how beta moves. This is why methods that can learn how beta moves from a wider set of inputs are useful.

2.4 Machine Learning in Asset Pricing and Beta Forecasting

Machine learning has changed empirical asset pricing over the past decade. Gu et al. (2020) compare many methods for forecasting returns and risk premia and find that neural networks and tree-based models clearly do better than linear regressions, mainly because they capture nonlinear effects and interactions between predictors. This flexibility suits beta forecasting well, because a wide set of inputs may matter. Besides past betas, several firm characteristics have been linked to systematic risk or to expected returns, including firm size, illiquidity (Amihud, 2002), idiosyncratic volatility (Ang et al., 2006), and momentum (Jegadeesh and Titman, 1993). These variables form the predictor set used in this study. The closest work to this study is Drobetz et al. (2025), who apply machine learning to forecast market beta and show, using a value-weighted out-of-sample comparison, that machine-learning estimators can match or beat strong traditional benchmarks.

2.5 Long Short-Term Memory Networks

Recurrent neural networks are designed for sequential data, but standard versions find it hard to learn long-range patterns because their gradients vanish over long sequences. The Long Short-Term Memory network of Hochreiter and Schmidhuber (1997) deals with this using a memory cell controlled by an input gate and an output gate, and Gers et al. (2000) later added the forget gate that completes the standard design. These gates let the network keep or remove information over long periods. Fischer and Krauss (2018) apply the LSTM to financial time series and find that it does better than models without memory, such as random forests and standard feed-forward networks, because its memory lets it use information from far back in the sequence and extract signal from noisy data. The LSTM has since been used widely in financial forecasting, including to predict a stock-market index (Bhandari et al., 2022). Because deep networks are sensitive to their random starting values, a common approach is to train several models from different starting points and average their forecasts. This is known as an ensemble, and it reduces variance (Hansen and Salamon, 1990). These features make the LSTM a good choice for forecasting a beta that changes over time.

2.6 Asset Pricing and Beta in the Thai Market

Research on the Thai market is fairly limited, although a few studies show that systematic risk on the Stock Exchange of Thailand changes over time. Prukumpai (2015) models industry-portfolio betas with a two-regime Markov-switching model and finds that they are unstable over time and depend on the regime, so that a constant-beta CAPM measures risk poorly. The same study compares these betas with a three-year rolling regression, which confirms the instability. Wider work on Thai asset pricing, such as Charoenwong et al. (2021), studies factor and anomaly pricing with a q-factor approach, but it does not deal with estimating beta for individual stocks. This existing work on Thai beta stays at the industry-portfolio level and uses parametric models. To date, no published study appears to have used a recurrent neural network to forecast systematic risk for individual SET100 stocks. This study addresses that gap by applying the machine-learning beta-forecasting approach to an emerging market, and by testing it against the benchmarks that practitioners actually use.

Chapter 3

Data

3.1 Sample and Data Sources

This study analyses the constituents of the SET100 Index, which comprises the 100 large-capitalization stocks listed on the Stock Exchange of Thailand (SET). The sample is free of survivorship bias and includes every stock that was a member of the SET100 in any month during the sample period with each stock entering or leaving the panel on the calendar date it joined or left the index. The sample period runs from April 2008 to December 2024, a panel of 201 month-ends. Out-of-sample evaluation covers the ten years from January 2015 to December 2024, a total of 120 forecast months.

Stock-level price and fundamental data are obtained from Refinitiv Workspace. The closing prices are adjusted for both capital actions and cash dividends. The market portfolio is proxied by the SET100 Total Return Index (TRI), and the risk-free rate by the Thai Government Bond 1 to 3 Year Total Return Index reported by the Thai Bond Market Association. The default spread is the yield differential between Moody's seasoned Baa-rated and Aaa-rated U.S. corporate bonds, sourced from the Federal Reserve Economic Data (FRED) database.

3.2 Return Measurement

Total stock return includes both capital gains and cash dividends. Since the closing prices are already adjusted for both capital actions and cash dividends, the daily continuously compounded total return is computed directly as follows,

$$r_{i,d} \;=\; \ln\!\left(\frac{P_{i,d}}{P_{i,d-1}}\right) \tag{1}$$

where $P_{i,d}$ is the adjusted closing price of stock i on trading day d. The daily excess return for both the stock and the market is then obtained by subtracting the daily risk-free continuously compounded return,

$$r^e_{i,d} \;=\; r_{i,d} - r_{f,d}. \tag{2}$$

The monthly continuously compounded excess return is constructed by summing the daily excess returns within each calendar month. For each stock i and month t,

$$r^e_{i,t} \;=\; \sum_{d=1}^{D} r^e_{i,d}, \tag{3}$$

where $r^e_{i,d}$ is the daily excess return on the d-th trading day of month t and D is the total number of trading days in month t.

3.3 Forecasting Target: Future Realized Beta

The forecasting target is the future market beta of a stock. Because the true beta is unobservable, this study follows Drobetz et al. (2025) and uses the realized beta computed from daily excess returns as the prediction target. The realized beta is a consistent estimator of a stock's average beta over the measurement window as the returns are sampled more finely (Andersen et al., 2006), and daily sampling over a one-year window is the standard implementation in this literature. For each stock i and month-end t, the realized beta over the forward k-month window is

$$\beta^{\mathrm{R}}_{i,t+k} = \frac{\mathrm{Cov}^{\mathrm{R}}_{iM,t+k}}{\mathrm{Var}^{\mathrm{R}}_{M,t+k}} \tag{4}$$

where $\mathrm{Cov}^{\mathrm{R}}_{iM,t+k}$ is the realized covariance of daily excess returns between stock i and the SET100 TRI over the k-month forward window and $\mathrm{Var}^{\mathrm{R}}_{M,t+k}$ is the realized variance of the market excess return computed in the same way over the same window. The horizon is set to $k = 12$ months, the most common choice in beta-forecasting applications.

3.4 Predictor Variables

Table 1 summarises the 10 predictor variables used to forecast future realized beta. The set follows the stock characteristics of Drobetz et al. (2025) and the firm-fundamental approach of Cosemans et al. (2016) for forecasting market beta.

Table 1: Variable Descriptions and Definitions

Table 1 presents the descriptions and definitions of the ten predictors used in this analysis. Stock-level price and fundamental data are obtained from Refinitiv Workspace. The three Fama-French daily factors are obtained from Thailand's Factor Library (Charoenwong et al., 2021), supported by the Thailand Capital Market Development Fund and the Stock Exchange of Thailand. The default spread is obtained from FRED. Accounting data are released with a minimum lag of four months from the fiscal-quarter end to avoid look-ahead bias.

#	Predictor	Description	Definition
Predictors Based on Historical Beta from Sample Estimates			
1	OLS_3M_D	Short-term beta	Sample beta estimate from rolling OLS using a 3-month window of daily excess returns
2	OLS_1Y_D	Medium-term beta	Sample beta estimate from rolling OLS using a 1-year window of daily excess returns
3	OLS_5Y_M	Long-term beta	Sample beta estimate from rolling OLS using a 5-year window of monthly excess returns
Predictors Based on Accounting Information			
4	ME	Firm size	Log market value of equity
5	DIVPAY	Dividend payout	Ratio of trailing-twelve-month cash dividends to net income
Predictors Based on Technical and Liquidity Indicators			
6	TO	Turnover	Log total monthly trading value
7	MOM	Momentum	Cumulative excess return from month $t-12$ to month $t-1$
8	IVOL	Idiosyncratic volatility	Log standard deviation of daily residuals from the Fama and French (1993) three-factor model within the current calendar month
9	ILLIQ	Illiquidity	Amihud (2002) ratio of monthly absolute return to monthly trading value
Predictor Based on Macroeconomic Indicators			
10	DFY	Default spread	Yield differential between Moody's seasoned Baa- and Aaa-rated U.S. corporate bonds

3.4.1 Historical Betas

The historical beta is obtained by running a time-series ordinary least squares (OLS) regression. For each stock i and month-end t, the historical beta is calculated as

$$r^e_{i,\tau} \;=\; \alpha^{\text{Hist}}_{i,t} + \beta^{\text{Hist}}_{i,t} \cdot r^e_{M,\tau} + \varepsilon_{i,\tau} \tag{5}$$

where $r^e_{i,\tau}$ and $r^e_{M,\tau}$ are the excess returns on stock i and the SET100 TRI over the rolling window τ ranging over the past observations ending at month-end t.

Following Drobetz et al. (2025), three sets of rolling betas from different window lengths and data frequencies are selected to feed into the LSTM with short-, medium-, and long-horizon information on the variation in beta. The first is the short-term beta (OLS_3M_D) using a 3-month historical window of daily excess returns. The second is the medium-term beta (OLS_1Y_D) using a one-year historical window of daily excess returns. The last is the long-term beta (OLS_5Y_M) using a 5-year historical window of monthly excess returns. Combining windows of different lengths lets the model draw on both recent and persistent variation in beta.

3.4.2 Accounting Information

Two predictors summarise firm fundamentals from accounting and market data: firm size and dividend payout.

Firm size is one of the principal firm characteristics, captured here by the market value of equity. For each stock i at month-end t, firm size is calculated as the logarithm of the firm's market capitalization,

$$\text{ME}_{i,t} \;=\; \ln\!\Big(P^{\text{close}}_{i,t} \cdot S_{i,t}/10^6\Big) \tag{6}$$

where $P^{\text{close}}_{i,t}$ is the closing price in baht and $S_{i,t}$ is the number of shares outstanding at month-end t. The logarithmic transformation reduces the strong right-skewness of the cross-sectional market-capitalization distribution.

Dividend payout proxies for a firm's maturity and financial stability. For each stock i at month-end t, the dividend payout ratio is calculated as

$$\text{DIVPAY}_{i,t} \;=\; \frac{D_{i,t}}{\text{NI}_{i,t}} \tag{7}$$

where $D_{i,t}$ is the sum of trailing-twelve-month cash dividends and $\text{NI}_{i,t}$ is the trailing-twelve-month net income of stock i, summed from the most recent four fiscal-quarter reports available at month-end t. To avoid look-ahead bias, accounting data are released with a minimum lag of four months from the fiscal-quarter end.

3.4.3 Technical and Liquidity Indicators

Four predictors capture trading activity, liquidity, and recent return behaviour: turnover, momentum, idiosyncratic volatility, and illiquidity.

Turnover proxies trading activity and liquidity. For each stock i at month-end t, the turnover is calculated as the natural logarithm of the total monthly trading value,

$$\text{TO}_{i,t} \;=\; \ln\!\left(\sum_{d=1}^{D} v_{i,d}\right) \tag{8}$$

where $v_{i,d}$ is the daily total trading value of stock i (in million baht) on the d-th trading day of calendar month t, and D is the total number of trading days in month t.

Momentum is the cumulative past return underlying the return-continuation effect of Jegadeesh and Titman (1993). For each stock i at month-end t, momentum is calculated as the cumulative continuously compounded excess return over the twelve calendar months ending one month before t,

$$\text{MOM}_{i,t} \;=\; \sum_{m=t-12}^{t-1} r^{e}_{i,m}, \tag{9}$$

where $r^{e}_{i,m}$ is the monthly continuously compounded excess return. The most recent month is excluded to avoid contamination from the short-term reversal effect.

Idiosyncratic volatility captures firm-specific return risk that is unrelated to common factor exposures (Ang et al., 2006). For each stock i at month-end t, the Fama-French three-factor model is estimated on daily excess returns within the calendar month of t,

$$r^{e}_{i,d} \;=\; \alpha_{i,t} + \beta^{M}_{i,t} \cdot \mathrm{MKT}_d + \beta^{S}_{i,t} \cdot \mathrm{SMB}_d + \beta^{H}_{i,t} \cdot \mathrm{HML}_d + \varepsilon_{i,d}, \tag{10}$$

where MKT_d, SMB_d, and HML_d are the daily market, size, and value factors of Fama and French (1993), and $\varepsilon_{i,d}$ is the daily residual. These three factors are obtained from Thailand's Factor Library, constructed for the Thai market by Charoenwong et al. (2021). The idiosyncratic volatility is then

$$\mathrm{IVOL}_{i,t} \;=\; \ln\Big(\sigma(\varepsilon_{i,d})\Big), \tag{11}$$

where $\sigma(\varepsilon_{i,d})$ is computed across all trading days d in the calendar month of t.

Illiquidity captures the price impact per unit of trading volume (Amihud, 2002). For each stock i at month-end t, and following the monthly construction of Drobetz et al. (2025), illiquidity is calculated as the ratio of the monthly absolute return to the monthly trading value,

$$\mathrm{ILLIQ}_{i,t} \;=\; \frac{\left|\sum_{d=1}^{D} r_{i,d}\right|}{\sum_{d=1}^{D} v_{i,d}}, \tag{12}$$

where $r_{i,d}$ is the daily log return of stock i on the d-th trading day of calendar month t, $v_{i,d}$ is the corresponding daily trading value (in million baht), and D is the total number of trading days in month t.

3.4.4 Macroeconomic Indicator

The default spread is a macroeconomic indicator that serves here as a proxy for global credit-risk conditions affecting the Thai market. For each month t, the default spread is calculated as the differential between Moody's seasoned Baa-rated and Aaa-rated U.S. corporate bond yields,

$$\mathrm{DFY}_t \;=\; \mathrm{BAA}_t - \mathrm{AAA}_t, \tag{13}$$

where BAA_t and AAA_t are the monthly average yields on Moody's seasoned Baa-rated and Aaa-rated U.S. corporate bonds, obtained from the Federal Reserve Economic Data (FRED) database.

3.5 Sample Construction and Data Cleaning

Before training the model, the data are filtered at the end of each month, keeping an observation only if all the necessary inputs are present. The binding requirement is the five-year monthly beta, which is computed only when at least three years (36 monthly observations) of past returns are available within the five-year window. As a result, newly listed stocks are excluded until they have enough history.

The five-year history requirement removes by far the most observations, and together with the forward window required for the realized-beta target it accounts for almost all of those dropped. Requiring the remaining predictors to be complete removes less than one percent more, so the predictor distributions are essentially unchanged. The final clean dataset averages 89 of the 100 index members in a normal month and spans 225 different stocks over 201 month-ends.

3.6 Descriptive Statistics

Table 2 reports the summary statistics of the daily and monthly returns for the stocks in SET100 and for the SET100 Total Return Index.

Table 2: Descriptive Statistics of Daily and Monthly Returns

Table 2 presents the descriptive statistics of continuously compounded total returns over April 2008 to December 2024 across the SET100 member and index. T is the number of trading days in the Panel A and the number of months in the Panel B.

Series	T	Mean	Std	Min	Median	Max
Panel A: Daily						
SET100 Stocks	4,080	0.03%	1.23%	-13.42%	0.09%	8.46%
SET100 TRI	4,080	0.02%	1.25%	-12.69%	0.05%	8.79%
Risk-free rate	4,080	0.01%	0.05%	-0.34%	0.01%	0.56%
Panel B: Monthly						
SET100 Stocks	201	0.61%	6.80%	-42.53%	1.15%	22.25%
SET100 TRI	201	0.50%	5.87%	-37.69%	0.84%	18.72%
Risk-free rate	201	0.19%	0.36%	-1.09%	0.19%	2.57%

The SET100 stocks earn an average daily return of 0.03 % with a standard deviation of 1.23 %, ranging from −13.42 % to 8.46 %, and an average monthly return of 0.61 % with a standard deviation of 6.80 %. Over the same period, the SET100 Total Return Index earns an average daily return of 0.02 % with a standard deviation of 1.25 %, and an average monthly return of 0.50 % with a standard deviation of 5.87 %. The two series therefore behave similarly, with the constituent average earning a little more, consistent with the greater influence of smaller stocks in a simple average across the constituents. The index records its worst monthly return of −37.69 % in October 2008. As expected, the risk-free rate is far smaller and steadier, averaging 0.01 % daily and 0.19 % monthly with standard deviations of only 0.05 % and 0.36 %, in line with the steady returns on short-term Thai government bonds.

Table 3: Descriptive Statistics of the Target and Predictor Variables

Table 3 presents the descriptive statistics of the realized beta target and the ten predictor variables. The sample is the cleaned panel with a total of 17,887 stock-month observations.

Variable	Mean	Std	Min	Median	Max
Target Variable					
Realized Beta (β^{R})	0.9442	0.3773	−2.0505	0.9219	3.2181
Predictors Based on Historical Beta from Sample Estimates					
OLS_3M_D	0.9382	0.4888	−3.2649	0.9032	5.7707
OLS_1Y_D	0.9432	0.3674	−1.4974	0.9230	3.1643
OLS_5Y_M	1.0864	0.4118	−0.3757	1.0721	2.9025
Predictors Based on Accounting Information					
ME	10.7401	1.2931	5.6255	10.5941	14.4586
DIVPAY	0.5534	1.8974	−10.3519	0.4345	68.1305
Predictors Based on Technical and Liquidity Indicators					
TO	7.9338	1.5302	−0.8371	7.9903	12.0018
MOM	0.0350	0.4254	−2.8600	0.0333	2.3782
IVOL	−4.2779	0.4656	−8.8737	−4.2905	−1.7530
ILLIQ	0.0003	0.0052	0.0000	0.0000	0.3873
Predictor Based on Macroeconomic Indicators					
DFY	1.0551	0.4569	0.5500	0.9600	3.3800

The realized beta is the target variable. It has a mean of 0.94 and a standard deviation of 0.38. The average SET100 stock therefore carries a market exposure slightly below one, and the cross-section spans a wide range of risk profiles, from a handful of negative-beta stocks, through defensive low-beta names, to more cyclical high-beta stocks.

The three rolling-OLS historical betas have similar average values, between 0.94 and 1.09. The long-window monthly estimator (5-year monthly OLS beta) has the highest mean of 1.09, because long windows smooth out short-term changes and tend to carry the higher beta levels of earlier periods. The short-window daily estimator (3-month daily OLS beta) has the largest spread across stocks, with a standard deviation of 0.49 and the most extreme values from -3.26 to 5.77, which shows that short-window OLS estimates are very sensitive to recent shocks.

The two accounting predictors are measured on different scales. The log market value of equity has an average of 10.74 and a standard deviation of 1.29, which corresponds to a typical SET100 firm with a market capitalization of about 46 billion baht, while one standard deviation around this average covers roughly 13 billion to 168 billion baht. The dividend payout ratio is very spread out, with extreme values from -10.35 to 68.13 coming from firms with very small or negative net income. Its median of 0.43 shows that a typical SET100 firm pays about 43 % of its earnings as dividends, which points to a base of mature firms.

Among the technical and liquidity indicators, the log monthly trading value (turnover) has an average of 7.93, which is about 2.8 billion baht of monthly trading for a typical stock and matches the fact that the SET100 is the most liquid part of the Thai market. Momentum has a small positive mean (0.035), which shows that there is no strong average twelve-month trend across stocks. Log idiosyncratic volatility has a mean of -4.28, which corresponds to a daily stock-specific standard deviation of about 1.4 %, or an annualized idiosyncratic volatility of around 22 % for a typical stock. The Amihud illiquidity ratio is strongly skewed to the right, with most large values coming from small stocks with low trading volume, while the typical SET100 stock is very liquid, with a median close to zero.

The default spread, the single macroeconomic predictor, is the same for all stocks in any given month and ranges from 0.55 % to 3.38 % over the sample, with the highest values during the 2008 global financial crisis and the 2020 COVID-19 episode, when global credit risk increased sharply.

Chapter 4

Methodology

This chapter sets out how to test the two hypotheses of the study. The first hypothesis is that a Long Short-Term Memory (LSTM) network, trained on a panel of firm-level and macroeconomic predictors, forecasts the one-year-ahead realized beta of SET100 stocks more accurately than rolling-OLS benchmarks estimated from historical returns. The second hypothesis is that the LSTM's relative accuracy is stable across the pre- and post-COVID regimes. Two further analyses then deepen the accuracy comparison. One traces where the advantage lies across firm size, and the other asks which predictors drive the forecasts.

4.1 Data Preparation

4.1.1 Sliding-Window Sequences

The model is estimated on a sample of complete twelve-month input sequences. A forecast origin is drawn from the cleaned panel of Section 3.5, and a stock-month is kept as an origin only when two conditions hold. First, the origin month is a current SET100 member. Second, each of the twelve months in the lookback window is fully observed, with the realized-beta target and all ten predictors of Section 3.4 present and no missing cells. The ten predictors form the input, and the realized beta at the origin is the training label. Membership is required only at the origin, so a stock that briefly leaves the SET100 during the trailing year still contributes its predictor history for the interior months.

As a result, the panel yields 16,285 estimation sequences, spanning 208 distinct stocks with forecast origins from March 2009 to December 2024. For every forecast origin, the model receives an input matrix $(X_{i,t})$ that structures ten predictors over a twelve-month lookback window

$$X_{i,t} = \begin{bmatrix} x^1_{i,t-L+1} & x^2_{i,t-L+1} & \cdots & x^{10}_{i,t-L+1} \\ x^1_{i,t-L+2} & x^2_{i,t-L+2} & \cdots & x^{10}_{i,t-L+2} \\ \vdots & \vdots & \ddots & \vdots \\ x^1_{i,t} & x^2_{i,t} & \cdots & x^{10}_{i,t} \end{bmatrix} \tag{14}$$

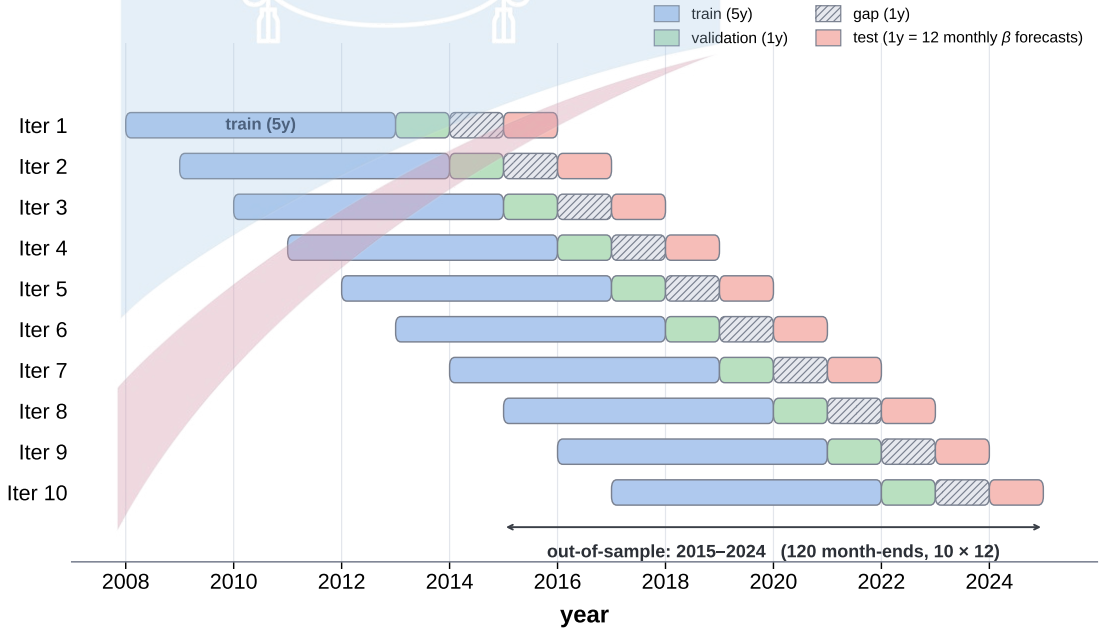
where the lookback length is $L = 12$ months and each row holds the ten predictors of Section 3.4 for one month. The target is the realized beta of Section 3.3 over the following $k = 12$ months.

$$y_{i,t} \;=\; \beta^{\mathrm{R}}_{i,t+k} \tag{15}$$

The sequences are split in time by a walk-forward design that uses no future information. Each iteration divides the panel into ordered blocks: five years for training, one year for validation to govern early stopping, a one-year gap, and one year for out-of-sample testing. The window slides forward one calendar year each time, and the ten iterations together produce monthly out-of-sample forecasts from January 2015 to December 2024 as shown in Figure 1. The model is trained once per iteration and then forecasts at each of the twelve month-ends of its test year, so the ten iterations together cover 120 out-of-sample month-ends. The LSTM is thus re-trained once a year, whereas the rolling-OLS benchmarks are re-estimated every month.

Figure 1: Walk-Forward Estimation and Forecasting Scheme

This figure reports the walk-forward scheme. Each row is one iteration, which divides the panel in time into five years for training, one year for validation, an unused one-year gap, and one year of out-of-sample testing; the window slides forward one year per iteration, and the ten test years together tile the out-of-sample period from January 2015 to December 2024.



4.1.2 Data Normalization

Before entering the network, the predictors are scaled to a common range from minus one to plus one. Following Gu et al. (2020), the nine firm-level predictors are rank-standardised each month across the cross-section, so that all features share a common scale. Each predictor is ranked over all stocks present that month, and the rank is mapped to this range,

$$\tilde{x}_{i,t}^{j} \;=\; 2\,\frac{r_{i,t}^{j}-1}{n_t-1} \;-\; 1, \tag{16}$$

where $r_{i,t}^{j}$ is the rank of predictor j for stock i over the n_t stocks that month, with ties averaged, so the smallest value maps to minus one, the largest to plus one, and the median to about zero. Because it is computed within each month, this transform carries no look-ahead. The macroeconomic series DFY takes the same value for every stock in a month and so cannot be ranked; it is instead min-max scaled to the same range,

$$\tilde{x}_{t}^{\mathrm{DFY}} \;=\; 2\,\frac{x_{t}^{\mathrm{DFY}}-x_{\min}^{\mathrm{DFY}}}{x_{\max}^{\mathrm{DFY}}-x_{\min}^{\mathrm{DFY}}} \;-\; 1, \tag{17}$$

where the minimum and maximum are taken from the same training set.

4.2 Forecasting Accuracy: LSTM versus Rolling-OLS Benchmarks

This section sets out the test of the first hypothesis, that the LSTM model forecasts the one-year-ahead realized beta more accurately than the rolling-OLS benchmarks. The LSTM model and the benchmarks are evaluated on the same out-of-sample window of a total of 120 month-end forecasts from January 2015 to December 2024.

4.2.1 Benchmark Estimators

The benchmark is the rolling-window OLS estimator, the standard approach to estimating a time-varying beta. It assumes that beta is persistent, so the forecast of next period's beta is simply the historical beta estimated over the most recent rolling window, which is re-estimated at every month-end.

The estimator draws only on the stock's own past returns and those of the market index. To estimate the historical beta for a stock at the end of a given month, the following regression is run over a fixed estimation window:

$$r^{e}_{i,t,s} \;=\; \alpha_{i,t} + \beta^{\text{Benchmark}}_{i,t}\, r^{e}_{M,t,s} + \varepsilon_{i,t,s} \tag{18}$$

where the dependent variable is the stock's excess return and the explanatory variable is the excess return of the SET100 TRI, both measured over the estimation window.

Following Drobetz et al. (2025), this study uses two variants that sit at opposite ends of the bias and variance trade-off typical of rolling regressions. The first is a long, low-frequency regression on five years of monthly returns (OLS_5Y_M), which gives a smooth estimate that adapts only gradually as beta shifts. The second is a short, high-frequency regression on one year of daily returns (OLS_1Y_D), which tracks recent movements far more quickly at the cost of more estimation noise. Both are conventional rolling-window benchmarks in standard use among practitioners.

4.2.2 LSTM Estimator

The LSTM network, introduced by Hochreiter and Schmidhuber (1997) and refined with a forget gate by Gers et al. (2000), is a recurrent neural network that reads a sequence one step at a time and uses a gated memory to hold the slow-moving part of beta over many months while still responding to recent movements, which simpler recurrent networks cannot do well.

To forecast the realized beta, the network is applied to the twelve-month input sequences. Its configuration is set out in Table 4 and is fixed in advance and held the same across all ten walk-forward iterations.

Table 4: LSTM Hyperparameter Configuration

This table provides the definitions and specifications of the LSTM hyperparameters, where the configuration is fixed in advance and held the same across all ten walk-forward iterations.

Hyperparameter	**Value**	**Definition**
Number of LSTM layers	1	How many times the sequence is re-encoded before the forecast; a single layer keeps the weight count within reach of the sample.
Hidden units per layer	8	Width of the LSTM layer, and so its memory capacity; kept narrow to prevent overfitting on the small sample.
Lookback length	12 months	Number of months in each input sequence, matching the one-year forecast horizon.
Dropout rate	0.1	Fraction of units randomly muted at each training step to curb overfitting.
Output activation	Linear	Maps the final hidden state to an unbounded continuous beta forecast.
Optimiser	RMSprop ($\eta = 0.001$)	Update rule that scales each step by the recent size of its gradients, suited to the time-varying nature of beta.
Batch size	128	Number of sequences used to compute each gradient update, balancing gradient stability against memory cost.
Maximum epochs	1,000	Ceiling on the passes over the training data, set high so that early stopping governs the halt.
Early-stopping patience	10 epochs	Training halts if the validation loss fails to improve for ten consecutive epochs, then restores the best weights.
Training loss	Value-weighted MSE	Fits the network on the same criterion used to evaluate it.
Ensemble size	10	Forecast averages ten networks trained from different random seeds.

The architecture is kept deliberately small, because each iteration trains on only about five thousand sequences and a wider or deeper network would carry more weights than the sample can identify. The network therefore uses a single LSTM layer of eight hidden units, following the evidence in Drobetz et al. (2025) that the accuracy of neural-network beta forecasts deteriorates as layers are added. The lookback is fixed at twelve months, matching the one-year forecast horizon and the trailing window over which membership and the predictors are required in Section 4.1.1, since the slow-moving part of beta is already carried by the five-year and one-year historical-beta predictors of Section 3.4. The remaining settings are reported in Table 4: the maximum number of epochs is set to a high ceiling of 1,000 so that early stopping, rather than a fixed schedule, ends training, and the network is fitted on the value-weighted mean squared error, the same criterion used to evaluate it.

The cell carries a running summary of the stock's recent history and updates it one month at a time. At each month τ in the twelve-month lookback window, the network reads that month's ten predictors, written x_τ, together with the summary built up to the previous month, written $h_{\tau-1}$, and decides what to forget, what to add, and what to pass on. Each decision is made by a gate, which applies the logistic sigmoid σ to turn its inputs into a number between zero and one; the gate then acts as a fraction that sets how much of a quantity passes through, from none at zero to all of it at one. The terms W and U and the biases b in the formulas below are numbers the network learns during training, and $\odot$ multiplies two vectors entry by entry.

The forget gate f_τ sets what fraction of the existing summary $c_{\tau-1}$ to keep. A value near zero drops information the network no longer needs, such as evidence from before a change in the stock's risk, while a value near one carries it forward unchanged,

$$f_\tau \;=\; \sigma(W_f x_\tau + U_f h_{\tau-1} + b_f)\,. \tag{19}$$

The input gate i_τ sets what fraction of new information to add. Alongside it, the candidate $\tilde{c}_\tau$ proposes that new information from this month's predictors, with the tanh function keeping the proposed values in a bounded range. The updated summary c_τ is then the part of the old summary that was kept plus the part of the new information that was added,

$$
\begin{aligned}
i_\tau &= \sigma(W_i x_\tau + U_i h_{\tau-1} + b_i)\,, \\
\tilde{c}_\tau &= \tanh(W_c x_\tau + U_c h_{\tau-1} + b_c)\,, \\
c_\tau &= f_\tau \odot c_{\tau-1} + i_\tau \odot \tilde{c}_\tau .
\end{aligned}
\tag{20}
$$

Finally, the output gate o_τ sets what fraction of the updated summary to report for the month. This reported value h_τ is passed on to the next month and, at the end of the window, used to form the forecast,

$$
\begin{aligned}
o_\tau &= \sigma(W_o x_\tau + U_o h_{\tau-1} + b_o)\,, \\
h_\tau &= o_\tau \odot \tanh(c_\tau)\,.
\end{aligned}
\tag{21}
$$

The summary is empty before the first month, so h and c start at zero, and these steps repeat month by month across the window. After all twelve months, the final summary h_t condenses the whole sequence into a single set of numbers. A simple linear formula turns this summary into the forecast of one trained network, weighting each number, adding them up, and adding a constant,

$$
\widehat{\beta}^{(s)}_{i,t+k|t} \;=\; W_{\text{out}}\, h_t + b_{\text{out}}
\tag{22}
$$

where W_{out} is a vector of weights and b_{out} a constant; the superscript s marks the forecast as that of a single network trained from random seed s. The gate weight matrices $\{W_f, W_i, W_o, W_c\}$ and $\{U_f, U_i, U_o, U_c\}$, the biases $\{b_f, b_i, b_o, b_c\}$, and the output parameters W_{out} and b_{out} are the trainable parameters of the network. They have no closed form and are estimated numerically, by minimising the value-weighted mean squared error between the forecast and the realized target,

$$
\text{MSE}_{\text{train}} \;=\; \sum_{(i,t)\,\in\,\text{train}} w_{i,t} \left(\beta^{\text{R}}_{i,t+k} - \widehat{\beta}^{(s)}_{i,t+k|t}\right)^2
\tag{23}
$$

where each training observation is weighted by its market-capitalization share within its origin month. The training loss is value-weighted so that the model is trained on the same criterion used to evaluate it.

To keep the result from depending on the random starting weights, the

network is trained $S = 10$ times from different seeds, and the reported forecast is the simple average of the S single-network forecasts,

$$\widehat{\beta}^{\mathrm{LSTM}}_{i,t+k|t} \;=\; \frac{1}{S}\sum_{s=1}^{S} \widehat{\beta}^{(s)}_{i,t+k|t} \tag{24}$$

following the ensemble convention of Hansen and Salamon (1990). This averaged forecast is the one evaluated in Section 4.2.3.

The trained ensemble is then held fixed for the rest of the iteration and used to forecast. Across its test year, every stock-month that meets the eligibility of Section 4.1.1, namely SET100 membership at the origin and a complete twelve-month predictor window, supplies that window to the ensemble, which produces the stock's forecast of the one-year-ahead realized beta.

4.2.3 Forecast Evaluation

The LSTM and the two rolling-OLS benchmarks differ in approach, but all three are evaluated on the same loss, the squared forecast error, aggregated at the end of each month as the value-weighted mean squared error,

$$\mathrm{MSE}_{t+k|t} \;=\; \sum_{i=1}^{N_t} w_{i,t}\left(\beta^{\mathrm{R}}_{i,t+k} - \widehat{\beta}^{m}_{i,t+k|t}\right)^2 \tag{25}$$

where N_t is the number of stocks in the sample at the end of month t, $w_{i,t}$ is the market-capitalization weight of stock i, and $\widehat{\beta}^{m}$ is the forecast of model m. The out-of-sample figure for a model is the average of these monthly values over the 120 month-ends, and a lower figure means more accurate forecasts. For the LSTM this matches the value-weighted squared-error criterion minimised in training (23), so the model is fitted and judged on the same measure. Because the value-weighted figure loads on the largest members, the same error and the test below are also reported under equal weighting, which replaces $w_{i,t}$ with $1/N_t$ so that every stock counts equally; comparing the two shows how far the aggregate result is driven by the large-cap stocks.

To test whether a difference in forecast error between the LSTM and a benchmark is statistically significant, this study follows Drobetz et al. (2025) and uses the Diebold and Mariano (1995) test of equal predictive accuracy, whose null

hypothesis is that the two models have equal expected loss. For each pair the difference in squared forecast errors at the stock level is taken and averaged across stocks with value weights to obtain the monthly loss differential,

$$d_t^{(\mathrm{LSTM},\, m_2)} \;=\; \sum_{i=1}^{N_t} w_{i,t} \Big[\Big(\beta^{\mathrm{R}}_{i,t+k} - \widehat{\beta}^{\mathrm{LSTM}}_{i,t+k|t} \Big)^2 - \Big(\beta^{\mathrm{R}}_{i,t+k} - \widehat{\beta}^{m_2}_{i,t+k|t} \Big)^2 \Big] \tag{26}$$

The differential is signed as the LSTM's squared error minus the benchmark's, so a negative value means the LSTM is the more accurate of the two. The DM statistic is the mean differential divided by its Newey and West (1987) standard error,

$$\mathrm{DM} \;=\; \frac{\bar{d}^{(\mathrm{LSTM},\, m_2)}}{\hat{\sigma}_{\bar{d}^{(\mathrm{LSTM},\, m_2)}}} \tag{27}$$

The DM test is a pairwise comparison, and its test statistic follows a standard normal distribution, so a negative DM statistic points to a lower LSTM loss, and the larger its magnitude, the more clearly the two models can be told apart. Because the forecasts overlap by twelve months, the monthly differentials follow a moving-average structure of order eleven, so the Newey–West lag is set to 11, one less than the forecast horizon (Diebold and Mariano, 1995; Harvey et al., 1997). With 120 monthly differentials in the full sample, the series is long enough for this asymptotic approximation, and the Newey–West variance already absorbs the overlap, so the statistic is referred to the standard normal without a small-sample adjustment.

4.3 Regime Stability across the COVID-19 Break

The accuracy comparison of Section 4.2.3 pools the whole out-of-sample period, which can mask a shift in the LSTM's standing around the COVID-19 shock. The study's second hypothesis is that its accuracy advantage is stable across this break rather than confined to one regime.

To examine this, the out-of-sample period is split at the forecast origin and the Diebold–Mariano test of Section 4.2.3 is re-run on the value-weighted loss within each part. The COVID-19 crisis window dated by Pisedtasalasai (2021) for

the Thai market, 17 January to 23 March 2020, is taken as the break itself, and the forecast origins that fall inside it are dropped.

Origins before the window form the pre-COVID sample and origins after it the post-COVID sample, so the two regimes are separated by the crisis rather than meeting at a single cut-off. Within each regime g the test statistic is

$$\mathrm{DM}_g \;=\; \frac{\bar{d}_g^{(\mathrm{LSTM},\, m_2)}}{\hat{\sigma}_{\bar{d}_g^{(\mathrm{LSTM},\, m_2)}}}, \qquad g \in \{\text{pre, post}\}, \tag{28}$$

where $\bar{d}_g$ is the mean monthly loss differential over the months in regime g, and its standard error is computed as in Section 4.2.3. A similar statistic in both regimes points to an advantage that holds through the break, whereas one that is significant before but not after, or the reverse, would mean the full-sample finding conceals a regime-dependent advantage.

4.4 Subsample Analysis across Size Terciles

The full-sample comparison of Section 4.2.3 gives one accuracy figure for the whole cross-section, which can hide differences across firm size. Because the value-weighted metric loads on the largest members, a model that forecasts large-cap betas well can dominate that figure even when it adds little for smaller stocks. To see whether the LSTM's accuracy is uniform across the size distribution, the comparison is repeated within size groups, following the size-sorted evaluation of Drobetz et al. (2025).

At each forecast origin the out-of-sample stocks are sorted into three groups by their market capitalization that month. Writing $r_{i,t}$ for the rank of stock i by market capitalization among the N_t stocks present at origin t, from smallest to largest, the size group is

$$g_{i,t} \;=\; \begin{cases} \text{small} & \text{if } r_{i,t} \le N_t/3, \\ \text{mid} & \text{if } N_t/3 < r_{i,t} \le 2N_t/3, \\ \text{large} & \text{if } r_{i,t} > 2N_t/3, \end{cases} \tag{29}$$

so each group holds about a third of that month's stocks, roughly twenty-eight.

Sorting at every origin keeps the groups dynamic, so a stock moves between terciles as its relative size changes. Within each tercile the value-weighted MSE of the LSTM and the two benchmarks is computed as in Section 4.2.3, the value-weighted figure using market-capitalization shares inside the group, and the Diebold–Mariano test of the LSTM against each benchmark is re-run on the within-group loss differential, keeping the Newey–West lag of eleven.

4.5 Predictor Importance

A neural network does not give a simple coefficient for each predictor the way a linear regression does. In order to answer which of the ten predictors the LSTM relies on, and whether only a few of them drive its forecasts rather than all ten mattering equally, each predictor's importance is measured by removing it and recording how much the forecasts worsen.

In the spirit of the zero-out method of Gu et al. (2020), who set each predictor to zero within the training sample and measure the fall in predictive fit, one predictor at a time is muted by replacing its value in every input sequence with its average, leaving the other nine unchanged. Fixing a predictor at a single value removes the differences across stocks that the model would otherwise use, so if the model depends on that predictor its forecasts deteriorate, and the size of that drop shows how much it relied on the predictor. Each predictor is therefore set to its average over the training data of the iteration in question, and the DFY series to its training average under the same min-max scaling.

The muted test set is then passed through the trained LSTM ensemble and the out-of-sample mean squared error is computed again. The importance of predictor j, written VI_j^w for variable importance, is how much this error rises against the baseline with no predictor muted,

$$\mathrm{VI}_j^w \;=\; \mathrm{MSE}_{\mathrm{muted},\,j} - \mathrm{MSE}_{\mathrm{baseline}} \tag{30}$$

where $\mathrm{MSE}_{\mathrm{baseline}}$ is the model's out-of-sample error under value weighting as in Section 4.2.3, and $\mathrm{MSE}_{\mathrm{muted},\,j}$ is the same error after predictor j is muted. A large positive value means muting the predictor hurts accuracy a lot, so the LSTM leans heavily on the differences across stocks in it and a negative value means the model does slightly better without it. Whether a predictor's importance is

statistically distinguishable from zero is assessed with the Diebold–Mariano test of Section 4.2.3, here comparing the full-model forecast against the muted-model forecast. Let $\widehat{\beta}^{(-j)}_{i,t+k|t}$ denote the ensemble forecast when predictor j is muted. The monthly value-weighted loss differential between this forecast and the baseline forecast is

$$d_t^j \;=\; \sum_{i=1}^{N_t} w_{i,t}\Big[\Big(\beta^{\mathrm{R}}_{i,t+k} - \widehat{\beta}^{(-j)}_{i,t+k|t}\Big)^2 - \Big(\beta^{\mathrm{R}}_{i,t+k} - \widehat{\beta}^{\mathrm{LSTM}}_{i,t+k|t}\Big)^2\Big] \tag{31}$$

Its mean over the 120 origins equals the importance VI^{w}_{j}, and the test statistic is this mean divided by its standard error,

$$\mathrm{DM}_j \;=\; \frac{\bar{d}^{\,j}}{\hat{\sigma}_{\bar{d}^{\,j}}} \tag{32}$$

The standard error $\hat{\sigma}_{\bar{d}^{\,j}}$ is the Newey and West (1987) standard error at a lag of eleven, as in Section 4.2.3, and DM_j is referred to the standard normal. Because the trained network is held fixed and only its input is muted, the two forecast series are taken as given, so the test compares forecasts rather than re-estimated nested models.

Chapter 5

Empirical Results

5.1 Forecast Accuracy: LSTM versus Rolling-OLS Benchmarks

This first step compares the three models on their out-of-sample forecasts of one-year-ahead realized beta. The section first examines how the three sets of forecasts are distributed and how they move over time. It then reports the average forecast errors, tests whether the differences between the models are statistically significant, and traces where the advantage lies across size groups.

5.1.1 Distribution and Dynamics of the Forecasts

Before turning to forecast errors, this subsection asks how the three forecasts behave: how each is distributed across the cross-section and how each moves with the realized beta over time.

Table 5 reports descriptive statistics for the future realized beta, which is the forward one-year beta from Equation (4), and its three forecasts are the LSTM of Equation (24) and the two rolling-window OLS benchmarks of Equation (18). The two benchmarks are OLS_1Y_D, estimated on one year of daily returns, and OLS_5Y_M, estimated on five years of monthly returns.

Table 5: Descriptive Statistics of the Realized Beta and the Three Forecasts

Each statistic is computed over the pooled sample of 10,145 out-of-sample stock-month observations. Std and Skew denote the standard deviation and the sample skewness.

Series	Mean	Std	Skew	Min	Max
Realized Beta (β^{R})	0.9753	0.3893	0.7465	−0.5451	3.2181
Beta Estimators					
LSTM	0.9427	0.1910	−0.4345	0.3275	1.4604
OLS_1Y_D	0.9771	0.3759	0.5282	−1.4974	3.1643
OLS_5Y_M	1.1241	0.4254	0.3493	−0.2704	2.5400

Figure 2 illustrates three features of the forecasts. The clearest is that the LSTM pulls its estimates toward the center of the cross-section. Its distribution is the tallest and narrowest of the four, with a standard deviation of 0.19, about half the 0.39 of the realized beta, and its forecasts stay within a narrow band, from 0.33 to 1.46, with almost no extreme values. At first sight this looks like the regression-to-the-mean idea behind the classical beta adjustments of Blume (1975) and Vasicek (1973). The LSTM, however, does not shrink in the same way. The Blume and Vasicek methods apply a linear adjustment that moves each beta toward a central value, which narrows the spread but largely keeps the overall shape of the distribution. The LSTM instead changes the shape. The realized beta is right-skewed (skewness 0.75) and has a long tail of high-beta stocks, whereas the LSTM forecast is mildly left-skewed (skewness -0.43) because it compresses the high-beta side far more than the low-beta side. Its shrinkage is therefore nonlinear and asymmetric, rather than the linear correction of the classical models.

The two rolling-OLS benchmarks differ in how closely they track the realized distribution. The one-year daily benchmark matches it most closely, where its distribution almost coincides with that of the realized beta. The five-year monthly benchmark, by contrast, is biased upward, with its distribution shifted to the right, so it overstates beta for most stocks.

Figure 2: Distribution of Realized Beta and the Three Forecasts

This figure reports the pooled distribution of the realized beta and the three forecasts over the out-of-sample period and the dotted vertical lines mark the means.

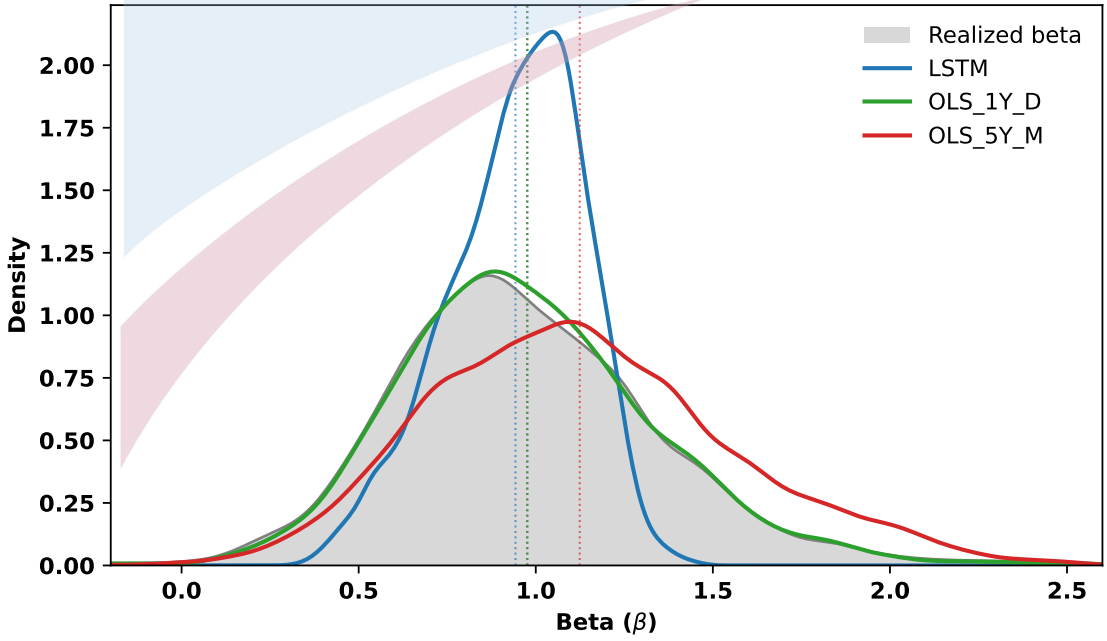


Table 6: Correlation with the Realized Beta, Pre- and Post-COVID

Pearson correlation of each forecast with the realized beta within each subsample, with the pre- and post-COVID split defined in Section 5.2 (4,982 pre-onset and 4,991 post-onset observations). Rank (Spearman) correlations give the same picture.

Forecast	Pre-COVID	Post-COVID
LSTM	0.49	0.57
OLS_1Y_D	0.49	0.58
OLS_5Y_M	0.27	0.29

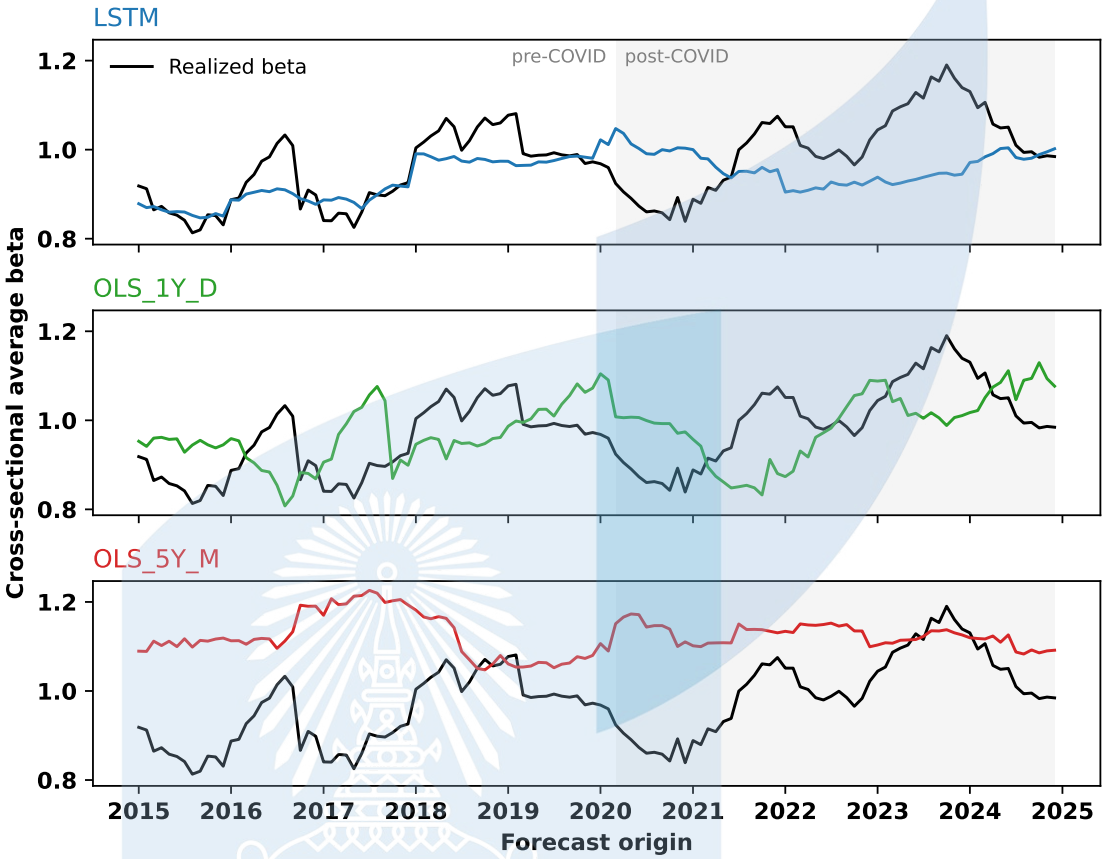
Table 6 turns from the shape of the forecasts to how closely each moves with the realized beta, reporting their correlation across the out-of-sample cross-section, with the sample split at the COVID-19 onset as defined in Section 5.2. The LSTM and the one-year daily benchmark track the realized beta about equally in both regimes, at 0.49 each before the onset and afterward at 0.57 and 0.58 respectively, while the five-year monthly benchmark trails, at 0.27 and 0.29. Co-movement therefore strengthens for all three after the onset without changing their order.

Figure 3 shows the dynamics behind these correlations, tracking the cross-sectional average beta from month to month. The realized average falls sharply at the 2020 onset, stays low through 2021 and 2022, and peaks in 2023. Each rolling benchmark carries the historical beta from its trailing window forward as next year's forecast. The one-year daily benchmark updates quickly and follows the realized series with only a short lag, whereas the five-year monthly benchmark adjusts slowly, missing the short-run moves and holding a level that stays above the realized series throughout the post-COVID fall.

The LSTM behaves differently. Because it is a learned forward predictor rather than a trailing estimate, it does not carry the past beta forward; instead, the cross-sectional compression of Figure 2 reappears over time as smaller movements around the average. It still updates with new information, unlike the slow five-year benchmark, but it responds less than the one-year benchmark to the movements in the realized beta. Across the panels of Figure 3, the realized beta and the two recent-window forecasts pass through the same turns, while the five-year benchmark stays high and nearly flat. This ordering anticipates the error ranking of the next section, where the two recent-window forecasts again stand close and the five-year benchmark is the weakest.

Figure 3: Cross-Sectional Average Beta over Time

This figure reports the cross-sectional average of the realized beta and each forecast in each out-of-sample month from January 2015 to December 2024. Each panel pairs one forecast (in color) with the realized beta (in black), and the shaded region marks the post-COVID regime.



5.1.2 Out-of-Sample Forecast Errors and Tests of Equal Predictive Accuracy

Tables 7 and 8 report the value-weighted and equal-weighted forecast accuracy of the three models over the 120 month-end forecasts from January 2015 to December 2024, computed from Equation (25), together with the Diebold–Mariano statistic of equal predictive accuracy between every pair of models from Equation (27).

Table 7: Out-of-Sample Forecast Accuracy (Value-Weighted)

The sample is the 120 month-end out-of-sample forecasts, January 2015 to December 2024. Panel A reports each model's average value-weighted MSE in percent over the $T = 120$ origins,

$$\mathrm{MSE}^{m}_{t+k|t} \;=\; \sum_{i=1}^{N_t} w_{i,t}\,\big(\beta^{\mathrm{R}}_{i,t+k} - \widehat{\beta}^{m}_{i,t+k|t}\big)^2, \qquad k = 12,$$

where N_t is the number of out-of-sample stocks in month t, $w_{i,t}$ the market-capitalization weight, $\beta^{\mathrm{R}}_{i,t+k}$ the realized beta and $\widehat{\beta}^{m}_{i,t+k|t}$ the model-m forecast, scaled by 100; bold marks the lowest error. Panel B reports the Diebold and Mariano (1995) test of the column model a against the row model b,

$$\mathrm{DM} \;=\; \frac{\bar{d}}{\hat{\sigma}_{\bar{d}}}, \qquad \bar{d} \;=\; \frac{1}{T}\sum_{t=1}^{T} d_t,$$

$$d_t \;=\; \sum_{i=1}^{N_t} w_{i,t}\left[\big(\beta^{\mathrm{R}}_{i,t+k} - \widehat{\beta}^{a}_{i,t+k|t}\big)^2 - \big(\beta^{\mathrm{R}}_{i,t+k} - \widehat{\beta}^{b}_{i,t+k|t}\big)^2\right],$$

with $\hat{\sigma}_{\bar{d}}$ the Newey and West (1987) standard error at a lag of eleven; a negative value means the column model has the lower loss.

Panel A. Average forecast errors

Model	**VW-MSE (%)**
LSTM	9.95
OLS_1Y_D	**9.74**
OLS_5Y_M	19.68
T (months)	120

Panel B. Equal predictive accuracy (Diebold–Mariano statistic)

	OLS_5Y_M	OLS_1Y_D	LSTM
vs. OLS_5Y_M	—	-4.33^{***}	-6.55^{***}
vs. OLS_1Y_D	4.33^{***}	—	0.20
vs. LSTM	6.55^{***}	-0.20	—

*, ** and *** denote significance at the 10%, 5% and 1% levels, respectively.

Table 8: Out-of-Sample Forecast Accuracy (Equal-Weighted)

The equal-weighted counterpart of Table 7, formed with equal weights $1/N_t$ in place of the market-capitalization weight $w_{i,t}$. Panel A reports each model's average equal-weighted MSE in percent over the $T = 120$ origins, scaled by 100; bold marks the lowest error. Panel B reports the Diebold and Mariano (1995) statistic of the column model against the row model, with the Newey and West (1987) standard error at a lag of eleven; a negative value means the column model has the lower loss.

Panel A. Average forecast errors

Model	**EW-MSE (%)**
LSTM	**10.94**
OLS_1Y_D	13.55
OLS_5Y_M	26.42
T (months)	120

Panel B. Equal predictive accuracy (Diebold–Mariano statistic)

	OLS_5Y_M	OLS_1Y_D	LSTM
vs. OLS_5Y_M	—	-6.52^{***}	-9.73^{***}
vs. OLS_1Y_D	6.52^{***}	—	-2.36^{**}
vs. LSTM	9.73^{***}	2.36^{**}	—

*, ** and *** denote significance at the 10%, 5% and 1% levels, respectively.

The results show how the three models rank on the value-weighted aggregate. Among the benchmarks, the five-year monthly estimator records the highest error, at 19.68%, and the one-year daily estimator the lowest, at 9.74%, while the LSTM lies between the two, at 9.95%. The Diebold–Mariano tests place these gaps statistically. Against the five-year monthly benchmark the statistic is -6.55 and significant at the 1% level, so the LSTM is the more accurate model by a clear and systematic margin. Against the one-year daily benchmark the statistic is 0.20 and far from significant, so the LSTM and the daily regression are statistically equal on the aggregate. The two benchmarks are themselves far apart: tested directly against each other, the one-year daily regression is the more accurate by a wide margin, with a Diebold–Mariano statistic of -4.33 value-weighted and -6.52 equal-weighted, both significant at the 1% level. The recent window is therefore the stronger benchmark whether or not the LSTM enters the comparison.

The equal-weighted results in Table 8 qualify this verdict. When every stock counts equally rather than by market capitalization, the LSTM records

the lowest error, at 10.94%, and improves on the one-year daily benchmark by a margin significant at the 5% level, with a Diebold–Mariano statistic of −2.36. The reversal between the two weightings is informative in itself. The advantage of the LSTM lies among the smaller stocks, so it shows through when stocks are weighted equally but is washed out under value weighting, where the large-cap stocks that the LSTM forecasts no better than the simple regression dominate the average. The size analysis of Section 5.1.3 traces this directly.

These patterns have clear reasons. The five-year monthly estimator is the weakest because a long estimation window adjusts slowly and carries a stale level into the forecast once beta moves. This reflects the estimation-window trade-off of Pesaran and Timmermann (2007). They show that the best window balances bias against variance, so that keeping some older data is often worthwhile, but when a large break occurs a long window which spans it carries stale information into the forecast. A slow five-year window cannot shed that stale level, so it loses accuracy when beta re-rates. That the LSTM clears this benchmark by a wide and significant margin suggests that the network recovers a more reliable beta from the wider predictor set than a slow rolling regression can. Unlike a single rolling regression, it can draw on several horizons of past beta at once, so it is not locked into the bias of any one window length. The one-year daily benchmark, by contrast, is already a strong forecast in its own right, because a year of daily returns tracks the recent level of beta closely and realized beta is predictable enough that this recent estimate carries forward well (Andersen et al., 2006; Becker et al., 2021). A flexible model of this kind adds the most when the estimate it competes against is unreliable, so where that estimate is already accurate the headroom to improve on it is correspondingly small. The near-tie on the aggregate is therefore not a sign that the wider predictor set is uninformative, but that much of its information is already embedded in a well-chosen recent window. The LSTM thus draws level with the one-year daily benchmark rather than beating it on the value-weighted aggregate. This is a more modest result than that of Drobetz et al. (2025), who find that machine learning estimators beat even strong rolling benchmarks, though their gains rest on random forests and a far larger predictor set than this study uses.

Overall, the LSTM is a dependable forecaster for the index as a whole, far ahead of the slow five-year benchmark and drawing level with the strong one-year daily benchmark on the value-weighted aggregate, while improving on it once every stock is weighted equally.

5.1.3 Forecast Accuracy across Size Groups

Tables 9 and 10 report the out-of-sample forecast accuracy of the three models within size terciles by market capitalization, under value and equal weighting.

Table 9: Forecast Accuracy by Size Group (Value-Weighted)

Out-of-sample stocks are sorted into size terciles by market capitalization in each origin month, about 28 stocks per group over the $T = 120$ origins. Panel A reports each model's average value-weighted MSE in percent within the group; bold marks the lowest error in each group. Panels B to D report, for each size group, the Diebold and Mariano (1995) statistic of the column model against the row model, defined as in Table 7; a negative value means the column model has the lower loss.

Panel A. Average forecast errors (VW-MSE, %)

Model	**Small**	**Mid**	**Large**
LSTM	**12.5**	**10.6**	9.6
OLS_1Y_D	16.7	13.7	**8.5**
OLS_5Y_M	36.1	22.8	18.0

Panel B. Small group — Diebold–Mariano statistic

	OLS_5Y_M	OLS_1Y_D	LSTM
vs. OLS_5Y_M	—	−5.72***	−7.25***
vs. OLS_1Y_D	5.72***	—	−2.15**
vs. LSTM	7.25***	2.15**	—

Panel C. Mid group — Diebold–Mariano statistic

	OLS_5Y_M	OLS_1Y_D	LSTM
vs. OLS_5Y_M	—	−4.37***	−7.14***
vs. OLS_1Y_D	4.37***	—	−3.07***
vs. LSTM	7.14***	3.07***	—

Panel D. Large group — Diebold–Mariano statistic

	OLS_5Y_M	OLS_1Y_D	LSTM
vs. OLS_5Y_M	—	−3.47***	−4.36***
vs. OLS_1Y_D	3.47***	—	0.91
vs. LSTM	4.36***	−0.91	—

*, ** and *** denote significance at the 10%, 5% and 1% levels, respectively.

Table 10: Forecast Accuracy by Size Group (Equal-Weighted)

The equal-weighted counterpart of Table 9, formed with equal weights $1/N_t$ in place of the market-capitalization weight. Panel A reports each model's average equal-weighted MSE in percent within the group; bold marks the lowest error in each group. Panels B to D report, for each size group, the Diebold and Mariano (1995) statistic of the column model against the row model; a negative value means the column model has the lower loss.

Panel A. Average forecast errors (EW-MSE, %)

Model	**Small**	**Mid**	**Large**
LSTM	**13.0**	**10.9**	**8.9**
OLS_1Y_D	17.3	14.3	9.1
OLS_5Y_M	36.7	25.8	16.8

Panel B. Small group — Diebold–Mariano statistic

	OLS_5Y_M	OLS_1Y_D	LSTM
vs. OLS_5Y_M	—	-5.84^{***}	-7.37^{***}
vs. OLS_1Y_D	5.84^{***}	—	-2.10^{**}
vs. LSTM	7.37^{***}	2.10^{**}	—

Panel C. Mid group — Diebold–Mariano statistic

	OLS_5Y_M	OLS_1Y_D	LSTM
vs. OLS_5Y_M	—	-4.37^{***}	-6.20^{***}
vs. OLS_1Y_D	4.37^{***}	—	-3.27^{***}
vs. LSTM	6.20^{***}	3.27^{***}	—

Panel D. Large group — Diebold–Mariano statistic

	OLS_5Y_M	OLS_1Y_D	LSTM
vs. OLS_5Y_M	—	-4.34^{***}	-5.49^{***}
vs. OLS_1Y_D	4.34^{***}	—	-0.26
vs. LSTM	5.49^{***}	0.26	—

*, ** and *** denote significance at the 10%, 5% and 1% levels, respectively.

The results show that the standing of the LSTM depends on the benchmark and on firm size. Against the five-year monthly benchmark the LSTM records the lower error in every group, with Diebold–Mariano statistics of -4 or lower and significant at the 1% level throughout, so the slow benchmark is dominated everywhere. Against the one-year daily benchmark the comparison turns on size.

Among small stocks the LSTM carries the lower error, at 12.5%, and the statistic of −2.15 is significant at the 5% level, so its lead is clear. Among mid-sized stocks it again carries the lower error, at 10.6%, and the statistic of −3.07 is significant at the 1% level, so the lead is clearest there. Among large stocks the order reverses, with the daily benchmark the more accurate model, but the statistic of 0.91 is far from significant, so the two are statistically equal.

This size pattern shows where the LSTM adds value. It earns its edge at the harder-to-estimate end of the cross-section, where a short return history gives a noisy beta, and adds nothing among the large stocks, where that history already gives a precise one. The pattern is consistent with Drobetz et al. (2025), who report that the gains from machine learning beta estimators are largest for smaller, less liquid stocks, and with Hollstein et al. (2019), who find that beta is hardest to forecast among smaller and less liquid names, which is where a better method has the most room to add value. The mechanism is shrinkage: trained to minimize squared error over a noisy panel, the network pulls the extreme estimates that simple regressions produce for thinly traded small stocks back toward the center of the cross-section, the regression-to-the-mean tendency that Blume (1975) and Vasicek (1973) build into classical beta adjustment.

Table 10 repeats this size-group comparison under equal weighting, with every stock counted equally rather than by market capitalization. The LSTM records the lowest error in all three groups, and its lead over the one-year daily benchmark is significant among small and mid-sized stocks, with Diebold–Mariano statistics of −2.10 and −3.27, while it stays insignificant among large stocks. The size pattern is therefore the same as under value weighting, with the cross-sectional advantage of the LSTM concentrated outside the large-cap segment.

Taken together, the analysis by size shows that the LSTM decisively outperforms the slow five-year monthly benchmark in every size group, and that it improves on the strong one-year daily benchmark among small and mid-sized stocks, at the 1% level in the mid group and the 5% level in the small group, while the two are statistically equal among the largest stocks. Its advantage is therefore concentrated outside the large-cap segment. Because the value-weighted aggregate is dominated by the largest stocks, where the LSTM holds no advantage, its edge among the smaller stocks does not show through at the index level, and the LSTM draws level with the one-year daily benchmark on the aggregate.

5.2 Regime Stability across the COVID-19 Break

The second step asks whether the relative accuracy of the LSTM is stable across the COVID-19 break. The out-of-sample period is split at the pandemic onset, following Section 4.3 where the crisis window follows Pisedtasalasai (2021).

5.2.1 Aggregate Regime Stability

Tables 11 and 12 report the out-of-sample forecast accuracy of the three models in the pre- and post-COVID subsamples, under value and equal weighting.

Table 11: Out-of-Sample Forecast Accuracy, Pre- and Post-COVID (Value-Weighted)

The out-of-sample period is split at the COVID-19 onset by forecast origin, using the crisis window of Pisedtasalasai (2021), 17 January to 23 March 2020, with the origins inside the window dropped, leaving 60 pre-onset and 58 post-onset origins. Panel A reports each model's average value-weighted MSE in percent within each subsample; bold marks the lowest error in each column. Panels B and C report, for the pre- and post-COVID subsamples, the Diebold and Mariano (1995) statistic of the column model against the row model, defined as in Table 7; a negative value means the column model has the lower loss.

Panel A. Average forecast errors (VW-MSE, %)

Model	**Pre-COVID**	**Post-COVID**
LSTM	**9.3**	10.7
OLS_1Y_D	9.4	**10.1**
OLS_5Y_M	16.5	23.1
T (months)	60	58

Panel B. Pre-COVID — Diebold–Mariano statistic

	OLS_5Y_M	OLS_1Y_D	LSTM
vs. OLS_5Y_M	—	-4.12^{***}	-7.84^{***}
vs. OLS_1Y_D	4.12^{***}	—	-0.07
vs. LSTM	7.84^{***}	0.07	—

Panel C. Post-COVID — Diebold–Mariano statistic

	OLS_5Y_M	OLS_1Y_D	LSTM
vs. OLS_5Y_M	—	-3.31^{***}	-5.27^{***}
vs. OLS_1Y_D	3.31^{***}	—	0.31
vs. LSTM	5.27^{***}	-0.31	—

*, ** and *** denote significance at the 10%, 5% and 1% levels, respectively.

Table 12: Out-of-Sample Forecast Accuracy, Pre- and Post-COVID (Equal-Weighted)

The equal-weighted counterpart of Table 11, formed with equal weights $1/N_t$ in place of the market-capitalization weight; the pre- and post-COVID subsamples (60 and 58 origins) are formed as there. Panel A reports each model's average equal-weighted MSE in percent; bold marks the lowest error in each column. Panels B and C report the Diebold and Mariano (1995) statistic of the column model against the row model; a negative value means the column model has the lower loss.

Panel A. Average forecast errors (EW-MSE, %)

Model	**Pre-COVID**	**Post-COVID**
LSTM	**9.7**	**12.4**
OLS_1Y_D	13.2	13.9
OLS_5Y_M	27.5	25.3
T (months)	60	58

Panel B. Pre-COVID — Diebold–Mariano statistic

	OLS_5Y_M	OLS_1Y_D	LSTM
vs. OLS_5Y_M	—	-6.57^{***}	-7.51^{***}
vs. OLS_1Y_D	6.57^{***}	—	-4.22^{***}
vs. LSTM	7.51^{***}	4.22^{***}	—

Panel C. Post-COVID — Diebold–Mariano statistic

	OLS_5Y_M	OLS_1Y_D	LSTM
vs. OLS_5Y_M	—	-3.47^{***}	-8.47^{***}
vs. OLS_1Y_D	3.47^{***}	—	-0.74
vs. LSTM	8.47^{***}	0.74	—

*, ** and *** denote significance at the 10%, 5% and 1% levels, respectively.

The results show that, against the five-year monthly benchmark, the LSTM records the lower error in both regimes, with a Diebold–Mariano statistic significant at the 1% level before the onset, at -7.84, and after it, at -5.27, so the slow benchmark is dominated in each subsample. Against the one-year daily benchmark the two are statistically level in both, with statistics of -0.07 before the onset and 0.31 after it, neither near significance. The error levels show which model the break hurts: across the onset the five-year benchmark's error rises sharply, while those of the LSTM and the one-year daily benchmark barely change.

The pattern suggests that the difference across the break comes from how quickly each model adjusts. The five-year monthly benchmark holds its estimate

over a long window, so it cannot keep up with the re-rating of betas after the onset, and its error rises. The LSTM and the one-year daily benchmark both use recent returns, so their forecasts adjust as the regime changes and their errors are little affected. This fits the estimation-window trade-off of Pesaran and Timmermann (2007): when a large break shifts the underlying relationship, a long window that spans it carries stale information into the forecast, so recent data become the better guide.

This result matters for the second hypothesis. A structural break is the setting in which a fixed long window is least able to track beta and an adaptive model is most useful, which suggests that the advantage of the LSTM over the long-window benchmark should be largest in such a period. The evidence supports this reading: that advantage persists across the break, and the gap in error behind it widens as the five-year benchmark deteriorates. The finding also coincides with Prukumpai (2015), who documents that systematic risk on the Stock Exchange of Thailand is time-varying and regime-dependent, indicating that a constant beta describes it poorly. The deterioration of the five-year benchmark across the COVID break is consistent with this view, while the LSTM and the one-year daily benchmark, which track recent betas, keep their accuracy.

Table 12 repeats the comparison under equal weighting, with every stock counted equally rather than by market capitalization. The LSTM holds the lowest error in both regimes, and its edge over the one-year daily benchmark is significant before the onset, with a Diebold–Mariano statistic of -4.22, but fades after it, at -0.74. This is the same pre-pandemic advantage that the equal-weighted aggregate shows, now visible once the large stocks no longer dominate the average.

Overall, the relative accuracy of the LSTM is stable across the COVID-19 break. It stays statistically equal to the one-year daily benchmark, while the gap in error to the five-year benchmark is, if anything, wider after the onset.

5.2.2 Regime Stability across Size Groups

Tables 13 and 14 report the out-of-sample forecast accuracy within size terciles for the pre- and post-COVID, under value and equal weighting.

Table 13: Forecast Accuracy by Size Group, Pre- and Post-COVID (Value-Weighted)

Stocks are sorted into size terciles in each origin month; the pre- and post-COVID subsamples are formed as in Table 11. Panel A reports each model's average value-weighted MSE in percent within each group-regime cell; bold marks the lowest error in each column. Panels B to D report, for each size group, the Diebold and Mariano (1995) statistic of the column model against the row model in the pre- and post-COVID subsamples, defined as in Table 7; a negative value means the column model has the lower loss.

Panel A. Average forecast errors (VW-MSE, %)

	Small		**Mid**		**Large**	
Model	Pre	Post	Pre	Post	Pre	Post
LSTM	**8.9**	**16.4**	**10.2**	**11.3**	9.2	10.0
OLS_1Y_D	15.1	18.3	14.4	13.1	**8.2**	**8.9**
OLS_5Y_M	38.1	34.0	25.0	20.6	13.7	22.6
T (months)	60	58	60	58	60	58

Panel B. Small group — Diebold–Mariano statistic

	Pre-COVID			**Post-COVID**		
	OLS_5Y_M	OLS_1Y_D	LSTM	OLS_5Y_M	OLS_1Y_D	LSTM
vs. OLS_5Y_M	—	-5.63^{***}	-6.09^{***}	—	-3.08^{***}	-7.38^{***}
vs. OLS_1Y_D	5.63^{***}	—	-5.67^{***}	3.08^{***}	—	-0.51
vs. LSTM	6.09^{***}	5.67^{***}	—	7.38^{***}	0.51	—

Panel C. Mid group — Diebold–Mariano statistic

	Pre-COVID			**Post-COVID**		
	OLS_5Y_M	OLS_1Y_D	LSTM	OLS_5Y_M	OLS_1Y_D	LSTM
vs. OLS_5Y_M	—	-3.48^{***}	-6.01^{***}	—	-2.67^{***}	-5.79^{***}
vs. OLS_1Y_D	3.48^{***}	—	-4.80^{***}	2.67^{***}	—	-1.07
vs. LSTM	6.01^{***}	4.80^{***}	—	5.79^{***}	1.07	—

Panel D. Large group — Diebold–Mariano statistic

	Pre-COVID			**Post-COVID**		
	OLS_5Y_M	OLS_1Y_D	LSTM	OLS_5Y_M	OLS_1Y_D	LSTM
vs. OLS_5Y_M	—	-2.69^{***}	-4.37^{***}	—	-3.11^{***}	-4.58^{***}
vs. OLS_1Y_D	2.69^{***}	—	0.75	3.11^{***}	—	0.60
vs. LSTM	4.37^{***}	-0.75	—	4.58^{***}	-0.60	—

*, ** and *** denote significance at the 10%, 5% and 1% levels, respectively.

Table 14: Forecast Accuracy by Size Group, Pre- and Post-COVID (Equal-Weighted)

The equal-weighted counterpart of Table 13, formed with equal weights $1/N_t$ in place of the market-capitalization weight. Panel A reports each model's average equal-weighted MSE in percent within each group-regime cell; bold marks the lowest error in each column. Panels B to D report, for each size group, the Diebold and Mariano (1995) statistic of the column model against the row model in the pre- and post-COVID subsamples; a negative value means the column model has the lower loss.

Panel A. Average forecast errors (EW-MSE, %)

	Small		**Mid**		**Large**	
Model	Pre	Post	Pre	Post	Pre	Post
LSTM	**9.8**	**16.5**	**10.1**	**12.0**	9.2	**8.7**
OLS_1Y_D	16.2	18.2	14.7	14.1	**8.8**	9.5
OLS_5Y_M	38.7	34.4	29.1	22.7	14.8	18.9
T (months)	60	58	60	58	60	58

Panel B. Small group — Diebold–Mariano statistic

	Pre-COVID			**Post-COVID**		
	OLS_5Y_M	OLS_1Y_D	LSTM	OLS_5Y_M	OLS_1Y_D	LSTM
vs. OLS_5Y_M	—	−6.47***	−6.49***	—	−2.94***	−6.34***
vs. OLS_1Y_D	6.47***	—	−4.79***	2.94***	—	−0.47
vs. LSTM	6.49***	4.79***	—	6.34***	0.47	—

Panel C. Mid group — Diebold–Mariano statistic

	Pre-COVID			**Post-COVID**		
	OLS_5Y_M	OLS_1Y_D	LSTM	OLS_5Y_M	OLS_1Y_D	LSTM
vs. OLS_5Y_M	—	−3.87***	−5.45***	—	−2.58**	−5.44***
vs. OLS_1Y_D	3.87***	—	−6.97***	2.58**	—	−1.11
vs. LSTM	5.45***	6.97***	—	5.44***	1.11	—

Panel D. Large group — Diebold–Mariano statistic

	Pre-COVID			**Post-COVID**		
	OLS_5Y_M	OLS_1Y_D	LSTM	OLS_5Y_M	OLS_1Y_D	LSTM
vs. OLS_5Y_M	—	−3.41***	−3.75***	—	−3.19***	−4.94***
vs. OLS_1Y_D	3.41***	—	0.29	3.19***	—	−0.64
vs. LSTM	3.75***	−0.29	—	4.94***	0.64	—

*, ** and *** denote significance at the 10%, 5% and 1% levels, respectively.

This further analysis asks whether the size-conditional standing of the LSTM, which Section 5.1.3 showed varies with firm size, is itself stable across the break.

The results show that, against the five-year monthly benchmark, the LSTM records the lower error in every group and every regime, with all Diebold–Mariano statistics significant at the 1% level, so the slow benchmark is dominated throughout. Against the one-year daily benchmark the picture changes across the break. Before the onset the LSTM carries the lower error among small and mid-sized stocks, with statistics of -5.67 and -4.80, both significant at the 1% level. After the onset these advantages turn insignificant, at -0.51 for small stocks and -1.07 for mid-sized stocks. Among large stocks the one-year daily benchmark is the more accurate model in both regimes, with statistics that are far from significant.

This pattern suggests that the cross-sectional edge of the LSTM is a pre-pandemic phenomenon. Before the onset the model is more accurate than the one-year daily benchmark among the small and mid-sized stocks, where a short return history makes beta hardest to estimate, while among large stocks, which the benchmark already estimates well, the two draw level. This is the segment in which Drobetz et al. (2025) report the gains from machine learning beta estimators to be largest. What the regime split adds is that the advantage is not permanent: after the onset it is no longer detectable. The most likely explanation is that the re-rating of betas after the onset raised the difficulty of forecasting in the smaller and less liquid names, where the advantage of the LSTM lay, which narrowed its lead over the simple daily regression.

Table 14 repeats the size-by-regime comparison under equal weighting. Before the onset the LSTM beats the one-year daily benchmark among small and mid-sized stocks, with Diebold–Mariano statistics of -4.79 and -6.97, both significant at the 1% level; after the onset these advantages turn insignificant. The equal-weighted result therefore mirrors the value-weighted one and confirms that the cross-sectional edge of the LSTM is a pre-pandemic feature that does not depend on the weighting.

Overall, the regime split shows that the relative standing of the three models is broadly stable across the COVID-19 break, with one qualification. The dominance of the LSTM over the slow long-window benchmark is robust everywhere, and the gap in error to that benchmark widens through the break, while its parity with the strong one-year daily benchmark holds on the value-weighted aggregate in both regimes. Beneath that aggregate, however, the edge of the LSTM among small and mid-sized stocks is a pre-pandemic feature that fades after the onset.

5.3 Predictor Importance

The third and final step opens the model and asks which of the ten predictors drive its forecasts, using the zero-out method of Section 4.5. The method scores each predictor by muting it, fixing it at its average value in the training data so it no longer varies across stocks, and measuring how much the out-of-sample error then rises. The ranking is computed over the full out-of-sample period.

Table 15 reports the value-weighted zero-out importance of each predictor, the increase in out-of-sample value-weighted MSE when that predictor is muted, together with the t-statistic of that increase.

Table 15: Predictor Importance

The importance VI_j^w of Equation (30) is the rise in out-of-sample value-weighted MSE when predictor j is muted to its training-fold mean,

$$\mathrm{VI}_j^w = \mathrm{MSE}_{\mathrm{muted},\,j} - \mathrm{MSE}_{\mathrm{baseline}}.$$

A large positive value means the model leans on the predictor; a negative value means the forecast is slightly better without it. The t-statistic tests VI_j^w against zero with the Newey and West (1987) standard error at a lag of eleven. Rows are ordered by importance, and the most important predictor is shown in bold.

Predictor	Importance	t-statistic
OLS_3M_D	**0.0144*****	**4.93**
OLS_1Y_D	**0.0110*****	**3.58**
IVOL	**0.0027***	**1.80**
TO	0.0014	1.20
OLS_5Y_M	0.0013	0.83
MOM	0.0010	0.61
DIVPAY	−0.0003	−0.39
DFY	−0.0005	−0.47
ILLIQ	−0.0007	−1.18
ME	**-0.0008***	**-1.74**

*, ** and *** denote significance at the 10%, 5% and 1% levels, respectively.

The ranking shows that a small core of predictors carries almost all of the signal, and the t-statistics confirm that only that core is statistically distinguishable from zero. The two short-horizon historical betas sit at the top, the three-month daily beta (OLS_3M_D) first at 0.0144 and the one-year daily beta (OLS_1Y_D) second at 0.0110, each significant at the 1% level. Idiosyncratic

volatility (IVOL) follows some way behind at 0.0027 and is significant only at the 10% level, while turnover (TO) at 0.0014 is not significant. The remaining predictors are statistically indistinguishable from zero: the five-year monthly beta (OLS_5Y_M) and momentum (MOM) sit just above zero at 0.0013 and 0.0010, and the dividend payout ratio (DIVPAY), the default spread (DFY) and illiquidity (ILLIQ) just below it. Size (ME) is the one characteristic with a marginally significant value, negative at -0.0008 at the 10% level, so the forecast is marginally better without it: the network places a small weight on size that adds more noise than signal. The effect is slight in both magnitude and significance, so it marks size as an uninformative input here rather than one that meaningfully harms the forecast. The signal is therefore concentrated in the two recent betas, and the network forecasts next year's beta mainly from this year's.

This pattern sits with the prior evidence. Drobetz et al. (2025) find that historical betas are the most informative input for a machine learning beta estimator, which matches the lead the two short-horizon betas take here. In their results turnover and size also rank near the top, whereas here turnover matters only a little and size not at all, plausibly because size and turnover vary far less across the large, liquid SET100 than across their broad cross-section. The heavy emphasis on the recent betas is consistent with the short-horizon predictability of realized beta documented by Andersen et al. (2006) and its long memory documented by Becker et al. (2021). That so few inputs carry the signal is in line with Drobetz et al. (2025) and the wider machine learning evidence of Gu et al. (2020), where a handful of variables usually dominate. The small role of the firm characteristics complements rather than contradicts Cosemans et al. (2016), who show that fundamentals-based priors improve linear beta forecasts: the network, fed three historical betas directly, recovers from them much of what those characteristics would add. It also agrees with Hollstein et al. (2019), who find for a broad cross-section that simple adjustments to a historical beta, namely exponential weighting and shrinkage toward a prior, forecast beta best, while more elaborate characteristic-based adjustments add little.

Recent realized betas already hold the cross-sectional information the model needs, so the network works mainly on combining them, not on the wider predictor set. Its preference for the short-horizon betas over the five-year beta echoes Section 5.2, where the slow long-window estimate forecast least well. Idiosyncratic volatility adds a little beyond the betas, but only at the 10% level, likely proxying for the liquidity frictions that bias a short-window beta.

Chapter 6

Conclusion

This study examined whether a Long Short-Term Memory model can forecast the one-year-ahead realized beta of SET100 stocks more accurately than the rolling-OLS estimators that practitioners rely on. Using a panel of ten firm-level and macroeconomic predictors, a walk-forward design that produces monthly out-of-sample forecasts from 2015 to 2024, and a value-weighted out-of-sample evaluation, it compared the network against a one-year daily and a five-year monthly rolling regression.

The evidence gives a conditional answer. First, on forecast accuracy, the LSTM does not uniformly beat the benchmarks. It draws level with the one-year daily regression on the value-weighted aggregate that the largest stocks dominate. Within the cross-section, however, it is the most accurate model for small and mid-sized stocks, significantly so in both groups over the full period, although this advantage rests on the pre-pandemic years and fades after the onset. It is edged out only among the largest names, and it beats the five-year monthly regression decisively in every size group. Second, on regime stability, the relative standing of the models is steady across the COVID-19 break: the lead over the five-year benchmark widens after the break while the parity with the one-year daily benchmark holds both before and after it, so the comparison reflects a structural property of the estimators rather than a feature of one market environment. Finally, on the drivers of the forecasts, a compact core of short-horizon historical betas explains most of what the network does, with idiosyncratic volatility adding a little more, while turnover, size and the default spread add no value.

More importantly, the central result runs against the common expectation that a flexible deep-learning model should dominate simple regressions across the board. A possible explanation for this finding is that for the large and liquid stocks that the value-weighted metric rewards, a one-year daily rolling regression already estimates beta with little bias and tracks its slow drift closely, so that little room is left for a richer model to add value. The network earns its advantage instead among the smaller and less liquid names, where a short return history makes the simple regression least reliable, and it does so chiefly through the flexible way it recombines the recent realized betas rather than through any extra signal drawn

from the wider predictor set.

The study makes two main contributions. It provides what appears to be the first application of a recurrent neural network to realized-beta forecasting in the Thai market, and it does so through a transparent value-weighted comparison against strong and well-established benchmarks rather than a single naive baseline. The practical implication follows directly from the size pattern: model complexity should be matched to the use case. For large-cap risk estimates a simple one-year daily rolling regression is hard to improve upon, so the additional cost of an LSTM is most justified for the small- and mid-cap segment, where its predictors genuinely help, and the importance results point practitioners towards short-horizon historical betas and idiosyncratic volatility as the most informative inputs for conditional beta.

Several limitations qualify these findings and point to future work. The SET100 cross-section is small relative to the United States samples used in the machine-learning beta literature, which limits the data available to train the network and may understate its potential. The architecture is deliberately parsimonious and is held fixed across iterations, so richer or adaptively tuned models might extract more signal. The analysis also covers a single market and a single beta definition. Future research could extend the universe to a broader pan-ASEAN sample, test attention-based or hybrid sequence models, consider alternative targets such as downside or conditional beta, and complement the statistical evaluation with an economic-value test, for example the performance of minimum-variance or anomaly-hedging portfolios formed on the competing beta forecasts. A final caveat is that the cross-sectional edge of the network is concentrated rather than uniform and, among the smaller names, is clearest before the pandemic, so its advantage should not be assumed to hold unchanged across every segment of the market or every market environment.

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